

Multi-Turn Speech Dialogue: Pre-2023 Supplement

A curated supplement of papers published in 2022 or earlier, excluding titles already listed in readme.docx.

Scope: multi-turn spoken interaction, spoken dialogue corpora and evaluation, turn-taking/interruption, incremental or barge-in-capable systems, and training methods for spoken dialogue. Text-only dialogue papers and single-turn ASR/TTS/SLU work are excluded unless they directly model turn-taking for spoken systems.

Cutoff	New papers	Deduplication
2022 and earlier	92	0 existing title(s) removed

Survey

1. **"Spoken Dialogue Technology: Enabling the Conversational User Interface"**. Michael F. McTear. ACM Computing Surveys 2002. [\[Paper\]](#)
2. **"POMDP-Based Statistical Spoken Dialog Systems: A Review"**. Steve Young et al.. Proceedings of the IEEE 2013. [\[Paper\]](#)
3. **"Using Neural Networks for Data-Driven Backchannel Prediction: A Survey on Input Features and Training Techniques"**. Markus Mueller et al.. HCI International 2015. [\[Paper\]](#)
4. **"A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version"**. Iulian V. Serban et al.. Dialogue & Discourse 2018. [\[Paper\]](#)
5. **"Turn-taking in Conversational Systems and Human-Robot Interaction: A Review"**. Gabriel Skantze. Computer Speech & Language 2021. [\[Paper\]](#)

General Surveys

1. **"A Survey on Dialogue Systems: Recent Advances and New Frontiers"**. Hongshen Chen et al.. SIGKDD Explorations 2017. [\[Paper\]](#)
2. **"Neural Approaches to Conversational AI"**. Jianfeng Gao et al.. Foundations and Trends in Information Retrieval 2019. [\[Paper\]](#)
3. **"Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey"**. Vevake Balaraman et al.. SIGDIAL 2021. [\[Paper\]](#)
4. **"Who Says What to Whom: A Survey of Multi-Party Conversations"**. Jia-Chen Gu et al.. IJCAI 2022. [\[Paper\]](#)

Datasets

1. **"The HCRC Map Task Corpus"**. Anne H. Anderson et al.. Language and Speech 1991. [\[Paper\]](#)
2. **"SWITCHBOARD: Telephone Speech Corpus for Research and Development"**. John J. Godfrey et al.. ICASSP 1992. [\[Paper\]](#)
3. **"The ICSI Meeting Corpus"**. Adam Janin et al.. ICASSP 2003. [\[Paper\]](#)
4. **"The Fisher Corpus: A Resource for the Next Generations of Speech-to-Text"**. Christopher Cieri et al.. LREC 2004. [\[Paper\]](#)
5. **"The AMI Meeting Corpus"**. Wessel Kraaij et al.. MLMI 2005. [\[Paper\]](#)
6. **"HKUST/MTS: A Very Large Scale Mandarin Telephone Speech Corpus"**. Yi Liu et al.. ISCSLP 2006. [\[Paper\]](#)
7. **"IEMOCAP: Interactive Emotional Dyadic Motion Capture Database"**. Carlos Busso et al.. Language Resources and Evaluation 2008. [\[Paper\]](#)

8. **"The SEMAINE Database: Annotated Multimodal Records of Emotionally Colored Conversations between a Person and a Limited Agent"**. Gary McKeown et al.. IEEE Transactions on Affective Computing 2012. [\[Paper\]](#)
9. **"The MAHNOB Mimicry Database: A Database of Naturalistic Human Interactions"**. Sanjay Bilakhia et al.. Pattern Recognition Letters 2015. [\[Paper\]](#)
10. **"The NoXi Database: Multimodal Recordings of Mediated Novice-Expert Interactions"**. Angelo Cafaro et al.. ICMI 2017. [\[Paper\]](#)
11. **"Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags"**. Koichiro Yoshino et al.. LREC 2018. [\[Paper\]](#)
12. **"Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset"**. Bill Byrne et al.. EMNLP-IJCNLP 2019. [\[Paper\]](#)
13. **"Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment"**. Benjamin Benk et al.. LREC 2020. [\[Paper\]](#)
14. **"HarperValleyBank: A Domain-Specific Spoken Dialog Corpus"**. Mike Wu et al.. arXiv 2020. [\[Paper\]](#)
15. **"The MSP-Conversation Corpus"**. Luz Martinez-Lucas et al.. Interspeech 2020. [\[Paper\]](#)
16. **"ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation"**. Holy Lovenia et al.. arXiv 2021. [\[Paper\]](#)
17. **"Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers"**. Michimasa Inaba et al.. LREC 2022. [\[Paper\]](#)
18. **"Design and Evaluation of the Corpus of Everyday Japanese Conversation"**. Hanae Koiso et al.. LREC 2022. [\[Paper\]](#)

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. **"Effects of System Barge-In Responses on User Impressions"**. Jun-ichi Hirasawa et al.. Eurospeech 1999. [\[Paper\]](#)
2. **"Learning Decision Trees to Determine Turn-Taking by Spoken Dialogue Systems"**. Ryo Sato et al.. ICSLP 2002. [\[Paper\]](#)
3. **"A Finite-State Turn-Taking Model for Spoken Dialog Systems"**. Antoine Raux and Maxine Eskenazi. NAACL-HLT 2009. [\[Paper\]](#)
4. **"Multimodal End-of-Turn Prediction in Multi-Party Meetings"**. Iwan de Kok and Dirk Heylen. ICMI-MLMI 2009. [\[Paper\]](#)
5. **"Evaluation and Optimisation of Incremental Processors"**. Okko Buss and David Schlangen. Dialogue & Discourse 2011. [\[Paper\]](#)
6. **"Turn-Taking Cues in Task-Oriented Dialogue"**. Agustin Gravano and Julia Hirschberg. Computer Speech & Language 2011. [\[Paper\]](#)
7. **"Continuously Predicting and Processing Barge-in During a Live Spoken Dialogue Task"**. Ethan O. Selfridge et al.. SIGDIAL 2013. [\[Paper\]](#)
8. **"Predicting User Satisfaction from Turn-Taking in Spoken Conversations"**. Shammur Absar Chowdhury et al.. Interspeech 2016. [\[Paper\]](#)
9. **"Towards a General, Continuous Model of Turn-Taking in Spoken Dialogue Using LSTM Recurrent Neural Networks"**. Gabriel Skantze. SIGDIAL 2017. [\[Paper\]](#)
10. **"Towards Deep End-of-Turn Prediction for Situated Spoken Dialogue Systems"**. Angelika Maier et al.. Interspeech 2017. [\[Paper\]](#)

11. **"Turn-Taking Estimation Model Based on Joint Embedding of Lexical and Prosodic Contents"**. *Chaoran Liu et al.*. Interspeech 2017. [\[Paper\]](#)
12. **"Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task"**. *Zakaria Aldeneh et al.*. ICASSP 2018. [\[Paper\]](#)
13. **"Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs"**. *Matthew Roddy et al.*. Interspeech 2018. [\[Paper\]](#)
14. **"Neural Dialogue Context Online End-of-Turn Detection"**. *Ryo Masumura et al.*. SIGDIAL 2018. [\[Paper\]](#)
15. **"Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers"**. *Kohei Hara et al.*. Interspeech 2018. [\[Paper\]](#)
16. **"Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network"**. *Nigel G. Ward et al.*. SLT 2018. [\[Paper\]](#)
17. **"An Incremental Turn-Taking Model for Task-Oriented Dialog Systems"**. *Andrei C. Coman et al.*. Interspeech 2019. [\[Paper\]](#)
18. **"Turn-Taking Prediction Based on Detection of Transition Relevance Place"**. *Kohei Hara et al.*. Interspeech 2019. [\[Paper\]](#)
19. **"Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions"**. *Daniel Ortega et al.*. ICASSP 2020. [\[Paper\]](#)
20. **"TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog"**. *Erik Ekstedt and Gabriel Skantze*. EMNLP Findings 2020. [\[Paper\]](#)
21. **"BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning"**. *Jin Yea Jang et al.*. EMNLP 2021. [\[Paper\]](#)
22. **"Projection of Turn Completion in Incremental Spoken Dialogue Systems"**. *Erik Ekstedt and Gabriel Skantze*. SIGDIAL 2021. [\[Paper\]](#)
23. **"Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation"**. *Shinya Fujie et al.*. Interspeech 2021. [\[Paper\]](#)
24. **"Contextual Acoustic Barge-In Classification for Spoken Dialog Systems"**. *Dhanush Bekal et al.*. Interspeech 2022. [\[Paper\]](#)
25. **"Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue"**. *Jiudong Yang et al.*. ICASSP 2022. [\[Paper\]](#)
26. **"Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation"**. *Jin Sakuma et al.*. Interspeech 2022. [\[Paper\]](#)
27. **"Turn-Taking Prediction for Natural Conversational Speech"**. *Shuo-Yiin Chang et al.*. Interspeech 2022. [\[Paper\]](#)
28. **"Voice Activity Projection: Self-Supervised Learning of Turn-Taking Events"**. *Erik Ekstedt and Gabriel Skantze*. Interspeech 2022. [\[Paper\]](#)

Evaluation Focus: Full-duplex Interaction

No additional eligible pre-2023 paper was found after deduplication. Early work in this area is represented under Half-duplex / Controlled Barge-in rather than modern native full-duplex interaction.

Evaluation Focus: Multi-turn Dialogue Capabilities

1. **"PARADISE: A Framework for Evaluating Spoken Dialogue Agents"**. *Marilyn A. Walker et al.*. ACL-EACL 1997. [\[Paper\]](#)
2. **"Quantitative and Qualitative Evaluation of DARPA Communicator Spoken Dialogue Systems"**. *Julie E. Boland et al.*. ACL 2001. [\[Paper\]](#)

3. **"The PARADISE Evaluation Framework: Issues and Findings"**. *Melita Hajdinjak and France Mihelic*. Computational Linguistics 2006. [\[Paper\]](#)
4. **"A Framework for Model-Based Evaluation of Spoken Dialog Systems"**. *Sebastian Moller et al.*. SIGDIAL 2008. [\[Paper\]](#)
5. **"User Simulation as Testing for Spoken Dialog Systems"**. *Hua Ai and Fuliang Weng*. SIGDIAL 2008. [\[Paper\]](#)
6. **"Spoken Dialog Challenge 2010: Comparison of Live and Control Test Results"**. *Alan W. Black et al.*. SIGDIAL 2011. [\[Paper\]](#)
7. **"Position Paper: Towards Standardized Metrics and Tools for Spoken and Multimodal Dialog System Evaluation"**. *Sebastian Moller et al.*. NAACL Workshop 2012. [\[Paper\]](#)
8. **"The Dialog State Tracking Challenge"**. *Jason D. Williams et al.*. SIGDIAL 2013. [\[Paper\]](#)
9. **"The Second Dialog State Tracking Challenge"**. *Matthew Henderson et al.*. SIGDIAL 2014. [\[Paper\]](#)

Models

Interaction Mode: Turn-taking

1. **"A Robust System for Natural Spoken Dialogue"**. *James F. Allen et al.*. ACL 1996. [\[Paper\]](#)
2. **"Galaxy-II: A Reference Architecture for Conversational System Development"**. *Stephanie Seneff et al.*. ICSLP 1998. [\[Paper\]](#)
3. **"Let's Go Public! Taking a Spoken Dialog System to the Real World"**. *Antoine Raux et al.*. Interspeech 2005. [\[Paper\]](#)
4. **"The RavenClaw Dialog Management Framework: Architecture and Systems"**. *Dan Bohus and Alexander I. Rudnicky*. Computer Speech & Language 2009. [\[Paper\]](#)
5. **"The SEMAINE API: Towards a Standards-Based Framework for Building Emotion-Oriented Systems"**. *Marc Schroder*. Advances in Human-Computer Interaction 2010. [\[Paper\]](#)
6. **"Attentive Listening System with Backchanneling, Response Generation and Flexible Turn-Taking"**. *Divesh Lala et al.*. SIGDIAL 2017. [\[Paper\]](#)
7. **"Towards End-to-End Spoken Dialogue Systems with Turn Embeddings"**. *Ali Orkan Bayer et al.*. Interspeech 2017. [\[Paper\]](#)

Interaction Mode: Half-duplex / Controlled Barge-in

1. **"A Multiparty Multimodal Architecture for Realtime Turntaking"**. *Kristinn R. Thorisson et al.*. IVA 2010. [\[Paper\]](#)
2. **"Towards Incremental Speech Generation in Dialogue Systems"**. *Gabriel Skantze and Anna Hjalmarsson*. SIGDIAL 2010. [\[Paper\]](#)
3. **"A General, Abstract Model of Incremental Dialogue Processing"**. *David Schlangen and Gabriel Skantze*. Dialogue & Discourse 2011. [\[Paper\]](#)
4. **"A Demonstration of Incremental Speech Understanding and Confidence Estimation in a Virtual Human Dialogue System"**. *David DeVault and David Traum*. SIGDIAL 2012. [\[Paper\]](#)
5. **"The InproTK 2012 Release"**. *Timo Baumann and David Schlangen*. NAACL Workshop 2012. [\[Paper\]](#)
6. **"An Easy Method to Make Dialogue Systems Incremental"**. *Hatim Khouzaimi et al.*. SIGDIAL 2014. [\[Paper\]](#)
7. **"Incremental Dialog Processing in a Task-Oriented Dialog"**. *Fabrizio Ghigi et al.*. Interspeech 2014. [\[Paper\]](#)

Interaction Mode: Full-duplex

No additional eligible pre-2023 paper was found after deduplication. Early work in this area is represented under Half-duplex / Controlled Barge-in rather than modern native full-duplex interaction.

Training Methods

Interaction Mode: Turn-taking

1. **"NJFun: A Reinforcement Learning Spoken Dialogue System"**. Diane Litman et al.. ANLP-NAACL Workshop 2000. [\[Paper\]](#)
2. **"Optimizing Dialogue Management with Reinforcement Learning: Experiments with the NJFun System"**. Satinder Singh et al.. Journal of Artificial Intelligence Research 2002. [\[Paper\]](#)
3. **"Agenda-Based User Simulation for Bootstrapping a POMDP Dialogue System"**. Jost Schatzmann et al.. HLT-NAACL 2007. [\[Paper\]](#)
4. **"Partially Observable Markov Decision Processes for Spoken Dialog Systems"**. Jason D. Williams and Steve Young. Computer Speech & Language 2007. [\[Paper\]](#)
5. **"Learning from Real Users: Rating Dialogue Success with Neural Networks for Reinforcement Learning in Spoken Dialogue Systems"**. Pei-Hao Su et al.. Interspeech 2015. [\[Paper\]](#)
6. **"A Sequence-to-Sequence Model for User Simulation in Spoken Dialogue Systems"**. Layla El Asri et al.. Interspeech 2016. [\[Paper\]](#)
7. **"End-to-End LSTM-Based Dialog Control Optimized with Supervised and Reinforcement Learning"**. Jason D. Williams and Geoffrey Zweig. arXiv 2016. [\[Paper\]](#)
8. **"On-Line Active Reward Learning for Policy Optimisation in Spoken Dialogue Systems"**. Pei-Hao Su et al.. ACL 2016. [\[Paper\]](#)
9. **"Deep Reinforcement Learning of Dialogue Policies with Less Weight Updates"**. Heriberto Cuayahuitl and Seunghak Yu. Interspeech 2017. [\[Paper\]](#)
10. **"Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations"**. Ruijie Yan et al.. arXiv 2022. [\[Paper\]](#)

Interaction Mode: Half-duplex / Streaming

1. **"Inverse Reinforcement Learning for Micro-Turn Management"**. Dongho Kim et al.. Interspeech 2014. [\[Paper\]](#)
2. **"An Incremental Turn-Taking Model with Active System Barge-In for Spoken Dialog Systems"**. Tiancheng Zhao et al.. SIGDIAL 2015. [\[Paper\]](#)
3. **"Reinforcement Learning for Turn-Taking Management in Incremental Spoken Dialogue Systems"**. Hatim Khouzaimi et al.. IJCAI 2016. [\[Paper\]](#)
4. **"A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning"**. Hatim Khouzaimi et al.. Computer Speech & Language 2018. [\[Paper\]](#)

Interaction Mode: Full-duplex

No additional eligible pre-2023 paper was found after deduplication. Early work in this area is represented under Half-duplex / Controlled Barge-in rather than modern native full-duplex interaction.